



Global trends in carbapenem- and difficult-to-treat-resistance among World Health Organization priority bacterial pathogens: ATLAS surveillance program 2018–2022

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ARTICLE INFO

Article history:

Received 28 February 2024

Accepted 28 March 2024

Available online 10 April 2024

Editor: Prof Wen-Chien Ko

Keywords:

Enterobacterales

Pseudomonas aeruginosa

Acinetobacter baumannii-calcoaceticus complex

Carbapenem resistance

Difficult-to-treat resistance

Surveillance

ABSTRACT

Objectives: To report trends in carbapenem resistance and difficult-to-treat resistance (DTR) among clinical isolates of Gram-negative priority pathogens collected by the ATLAS global surveillance program from 2018 to 2022.

Methods: Reference broth microdilution testing was performed in a central laboratory for 79,214 Enterobacterales, 30,504 *Pseudomonas aeruginosa*, and 13,500 *Acinetobacter baumannii-calcoaceticus* complex isolates collected by a constant set of 157 medical centres in 49 countries in Asia Pacific (APAC), Europe (EUR), Latin America (LATAM), Middle East-Africa (MEA), and North America (NA) regions. MICs were interpreted by 2023 CLSI M100 breakpoints. β -lactamase genes were identified for meropenem-nonsusceptible (MIC ≥ 2 mg/L) Enterobacterales isolates.

Results: Carbapenem-resistant Enterobacterales (CRE) detection increased ($P < 0.05$) in APAC, EUR, LATAM, and MEA regions and decreased in NA, while annual DTR percentages increased in all five regions. Carbapenem-resistant *P. aeruginosa* (CRPA; decreased in MEA region) and carbapenem-resistant *A. baumannii-calcoaceticus* complex (CRAB; decreased in MEA region and increased in EUR) remained relatively stable over time in all regions, although notably, annual percentages of CRAB and DTR *A. baumannii-calcoaceticus* complex isolates were consistently >25 percentage points lower in NA than in other regions. For all regions except NA, the majority of changes in CRE percentages could be attributed to hospital-acquired infections. Among meropenem-nonsusceptible Enterobacterales, KPC was the most frequent carbapenemase in NA and EUR each year. NDM was the most prevalent carbapenemase detected in 2022 in other global regions.

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Conclusion: CRE, CRPA, CRAB, and DTR rates vary among global regions over time highlighting the need for continuing surveillance to inform treatment strategies and antimicrobial stewardship.

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1. Introduction

Antimicrobial resistance is a threat to global public health [1,2]. It is associated with considerable patient morbidity and mortality as well as significant costs for both health systems and national economies. Antimicrobial resistance particularly impacts people living in low-resource settings and vulnerable populations. The emergence and spread of carbapenem-resistant Enterobacterales (CRE), *Pseudomonas aeruginosa* (CRPA), and *Acinetobacter baumannii-calcoaceticus* complex (CRAB), all among the World Health Organization's list of critical 'priority pathogens' is especially concerning given the importance of carbapenems as agents of last resort for patients infected with multidrug-resistant (MDR) Gram-negative pathogens [3]. Gram-negative bacilli exhibiting a difficult-to-treat resistance (DTR) phenotype, defined by non-susceptibility to all 'first-line' bactericidal antimicrobials used to treat serious infections (i.e., all β -lactams, including carbapenems, and fluoroquinolones), present clinicians with an even greater therapeutic challenge [4].

Non-susceptibility to carbapenems among Enterobacterales is most often mediated by the production of carbapenemases, although overexpression of ESBLs or AmpC β -lactamases in isolates with defective porins can play a role depending on the particular carbapenem [5]. The most widespread carbapenemases among the Enterobacterales are the Ambler class A serine carbapenemase, KPC, the class B metallo- β -lactamases (MBLs), including NDM, IMP, and VIM, and the class D OXA-48-like enzymes [6], although there is considerable variation in their geographical prevalence [7].

Global antimicrobial surveillance programs play a key role in our understanding of the emergence and spread of resistant bacterial pathogens. Data generated through such programs, whether publicly funded or private, can help to guide local treatment decisions, inform stewardship initiatives, and reveal gaps in the antimicrobial armamentarium that new agent development should target. ATLAS (Antimicrobial Testing Leadership and Surveillance) is a global surveillance program that annually monitors the in vitro activity of several antimicrobial agents against clinical isolates associated with bloodstream, intraabdominal, respiratory tract, skin/soft tissue, and urinary tract infections in countries in Asia/Pacific (APAC), Europe (EUR), Latin America (LATAM), Middle East/Africa (MEA), and North America (NA) regions [8].

This report examines resistance trends for the five global regions monitored by ATLAS, and one selected country from each region (Brazil, India, Spain, South Africa, and the United States) over a 5-y time-span from 2018 to 2022. For this analysis, we focused on carbapenem-resistance, as well as DTR phenotypes of Enterobacterales, *P. aeruginosa*, and *A. baumannii-calcoaceticus* complex. Regional 5-y trends for carbapenemase carriage by meropenem-nonsusceptible Enterobacterales were also analyzed.

2. Materials and methods

2.1. Bacterial isolates

The ATLAS global surveillance program requests that participating medical centres annually collect predefined numbers of

isolates of specific Gram-negative bacterial species from hospitalized patients with bloodstream, intra-abdominal, lower respiratory tract, skin and soft tissue, or urinary tract infections. Each isolate included in ATLAS is deemed a probable pathogen of a documented infection by a participating medical centre. The current analysis of ATLAS data only included isolates from medical centres that participated every year from 2018 to 2022. In total, 79,214 Enterobacterales, 30,504 *P. aeruginosa*, and 13,500 *A. baumannii-calcoaceticus* complex isolates from 157 medical centres in 49 countries were analyzed. Demographic information, including infection source, county, and year of collection, for the isolates included in this study is provided in Supplementary Table S1.

2.2. Isolate identification, antimicrobial susceptibility testing, and MIC interpretation

Bacterial isolates were shipped to a central laboratory (IHMA, Schaumburg, IL) where their species identification was confirmed using MALDI-TOF mass spectrometry (MALDI Biotyper®, Bruker Daltonics, Billerica, MA). Antimicrobial susceptibility testing was also performed by the central laboratory using the reference CLSI broth microdilution method [9] with MICs interpreted by 2023 CLSI M100 breakpoints [10]. CRE, CRPA, and CRAB were defined as isolates testing resistant by 2023 CLSI M100 criteria to imipenem and/or meropenem [10]. For the *Morganellaceae* family of Enterobacterales, only meropenem was considered in assigning resistance as this group of organisms intrinsically demonstrate reduced susceptibility to imipenem [10]. CRE, CRPA and CRAB rates were defined as the percentage of Enterobacterales, *Pseudomonas aeruginosa*, and *Acinetobacter baumannii-calcoaceticus* complex isolates testing carbapenem resistant. DTR was defined in accordance with the criteria proposed by Kadri, et al. [4] as an isolate testing nonsusceptible (intermediate or resistant) [10] to aztreonam, cefepime, ceftazidime, piperacillin-tazobactam, imipenem, meropenem, and levofloxacin. Aztreonam was not included in categorizing *A. baumannii-calcoaceticus* complex isolates as DTR because it is inactive against this group of species. Community-onset infections were defined as those present at hospital admission or within 48 h of admission, whereas presumptive nosocomial, or hospital-acquired infections, were defined as those manifesting 48 h or later after admission.

2.3. Molecular analysis

Isolates of Enterobacterales that tested with a meropenem MIC of ≥ 2 mg/L (i.e., meropenem-intermediate and -resistant) [10] were screened for the presence of genes encoding serine carbapenemases (KPC, OXA-48-like, and GES) and MBLs (NDM, IMP, VIM, SPM and GIM) using multiplex PCR assays, followed by amplification and sequencing of the full-length genes and comparison to publicly available databases, as previously described [11].

2.4. Statistical analysis

Trends in the annual percentage of Enterobacterales, *P. aeruginosa*, and *A. baumannii-calcoaceticus* complex isolates with

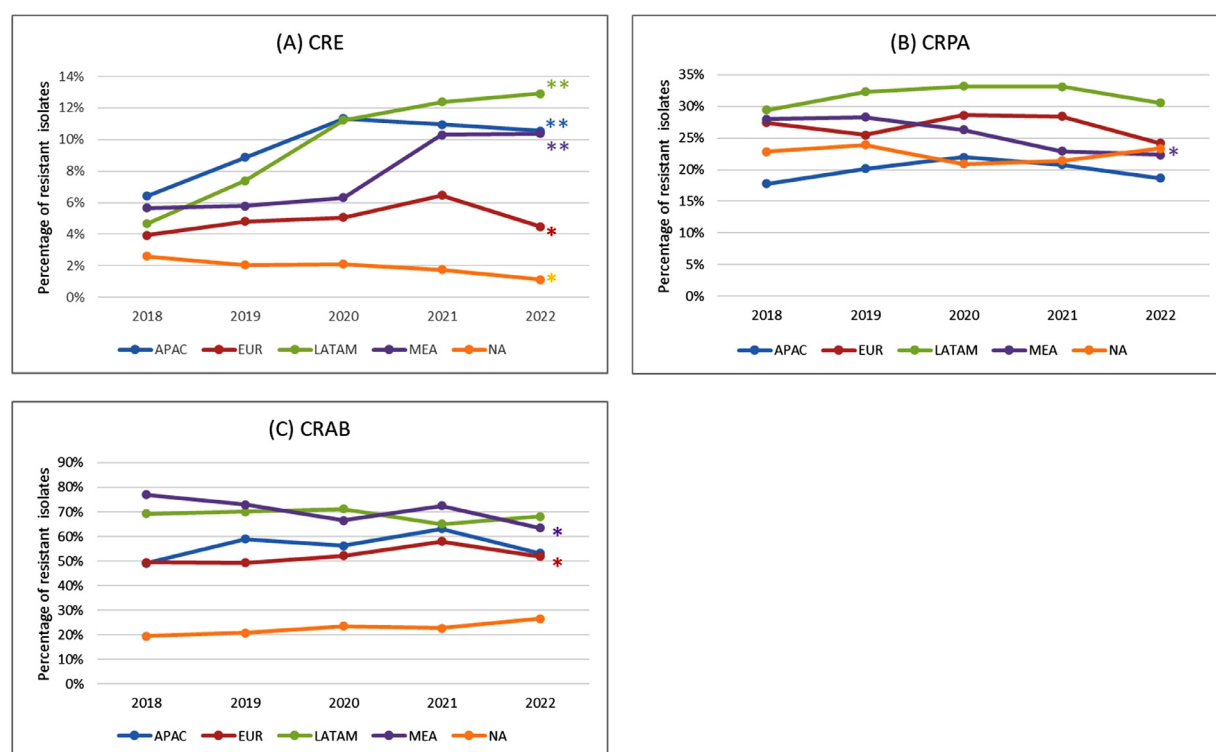


Fig. 1. Longitudinal resistance trends from 2018 to 2022 in the percentage of isolates identified as CRE (Panel A), CRPA (Panel B), and CRAB (Panel C) among all Enterobacteriales, *P. aeruginosa* and *A. baumannii-calcoaceticus* complex isolates by global region. Significance in trend over time determined by Cochran-Armitage test: ** $P < 0.001$, * $P < 0.05$. APAC, Asia/Pacific; EUR, Europe; LATAM, Latin America; MEA, Middle East/Africa; NA, North America. Percentages presented in this figure are provided in Supplementary Table S2.

resistant phenotypes from 2018 to 2022 were assessed for statistically significant differences using the Cochran-Armitage test for trend (JMP®, version 17, SAS, Cary, NC). A P -value < 0.05 was considered significant.

3. Results

3.1. Regional carbapenem resistance rates

Trends in annual carbapenem resistance rates among all Enterobacteriales, *P. aeruginosa*, and *A. baumannii-calcoaceticus* complex isolates over the 5-y period from 2018 to 2022 are shown in Figure 1. For Enterobacteriales, CRE detection increased significantly from 4.7% in 2018 to 12.9% in 2022 ($P < 0.0001$) for isolates from LATAM and from 5.7% in 2018 to 10.4% in 2022 ($P < 0.0001$) for isolates from MEA. For isolates from the APAC region, CRE prevalence peaked in 2020 (11.3%) and then remained relatively stable. The levels of CRE observed in isolates from EUR showed an increasing trend from 3.9% in 2018 to 6.5% in 2021 followed by a decrease to 4.5% in 2022 ($P = 0.0012$), whereas isolates from NA decreased in a step-wise manner from 2.6% in 2018 to 1.1% to 2022 ($P = 0.0012$). For *P. aeruginosa* isolates, a statistically significant trend in carbapenem resistance was only detected in the MEA region where step-wise decreases were observed from 28.0% in 2018 to 22.4% in 2022 ($P = 0.0078$). In each year, isolates from the LATAM region were more likely to be identified as CRPA than were isolates from other regions. Among *A. baumannii-calcoaceticus* complex isolates, CRAB annual percentages in EUR showed an increasing trend over the evaluation period ($P = 0.004$) and isolates from MEA showed a decreasing trend ($P = 0.0132$). Notably, CRAB rates were markedly lower in NA (by >25 percentage points) than in the other four regions each year.

3.2. Country-specific carbapenem resistance rates

Five countries, one in each global region (Brazil in the LATAM region; India in the APAC region; Spain in EUR; South Africa in the MEA region; and United States in NA), were selected to further investigate carbapenem resistance rates on a national level. Figure 2 presents trends in CRE, CRPA, and CRAB detection in these countries. In Brazil, significant increases in CRE and CRPA rates were observed, with CRE rates rising consistently each year in a linear fashion from 10.3% in 2018 to 35.6% in 2022 ($P < 0.0001$), and rates for CRPA increasing from 16.5% in 2018 to 55.6% in 2022 ($P < 0.0001$). In contrast, no clear trends were observed for CRE, CRPA, or CRAB for isolates collected in India, although it is noteworthy that the CRAB detection rate was high ($\geq 84\%$ each year). No clear trends in CRE, CRPA, or CRAB rates were detected among isolates from Spain over the studied time frame. In South Africa, a statistically significant trend of increasing CRE detection was noted from 4.4% in 2018 to 15.4% in 2022 ($P < 0.0001$), while CRPA and CRAB rates appeared to be decreasing; however, these changes were not statistically significant. In the United States, step-wise decreases in CRE were observed each year, from 2.8% in 2018 to 1.4% in 2022 ($P = 0.0008$), while CRAB rates increased linearly each year from 22.8% in 2018 to 33.3% in 2022 ($P = 0.0182$).

3.3. Regional CRE rates stratified by community-onset versus hospital-acquired infection

To determine whether carbapenem resistance was more often associated with community or nosocomial acquisition, CRE rates were determined separately for isolates from presumed community-onset infections (etiological agent isolated <48 h

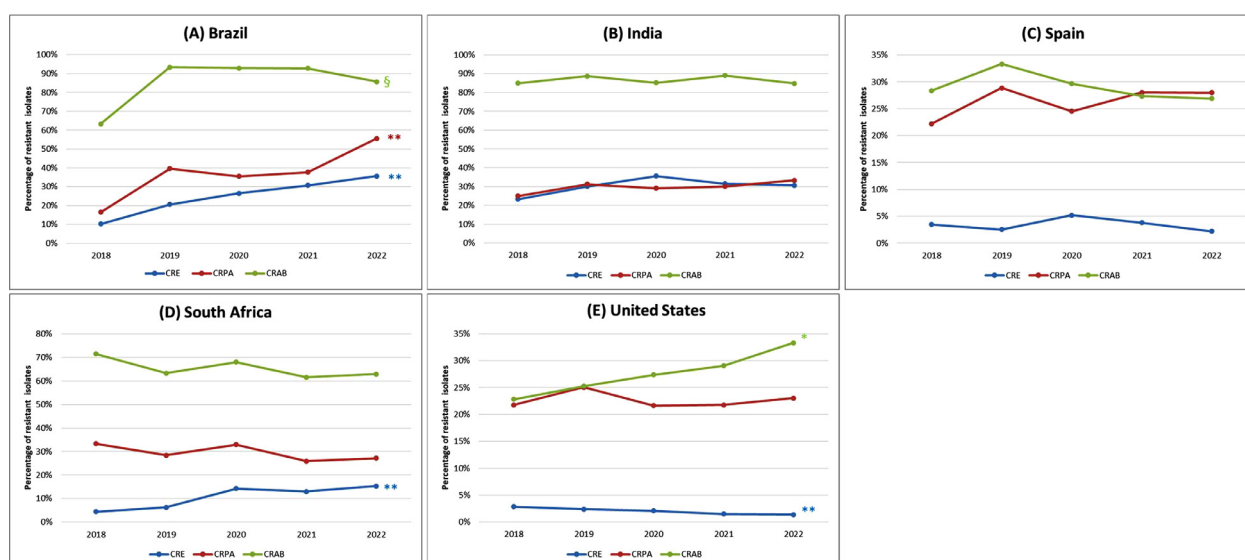


Fig. 2. Longitudinal resistance trends from 2018 to 2022 in the percentage of isolates identified as CRE, CRPA and CRAB among all Enterobacterales, *P. aeruginosa*, and *A. baumannii-calcoaceticus* complex isolates collected in Brazil (Panel A), India (Panel B), Spain (Panel C), South Africa (Panel D) and United States (Panel E). Significance in trend over time determined by Cochran-Armitage test: ** $P < 0.001$, * $P < 0.05$. §, Cochran-Armitage test invalid due to low n . Percentages presented in this figure are provided in Supplementary Table S3.

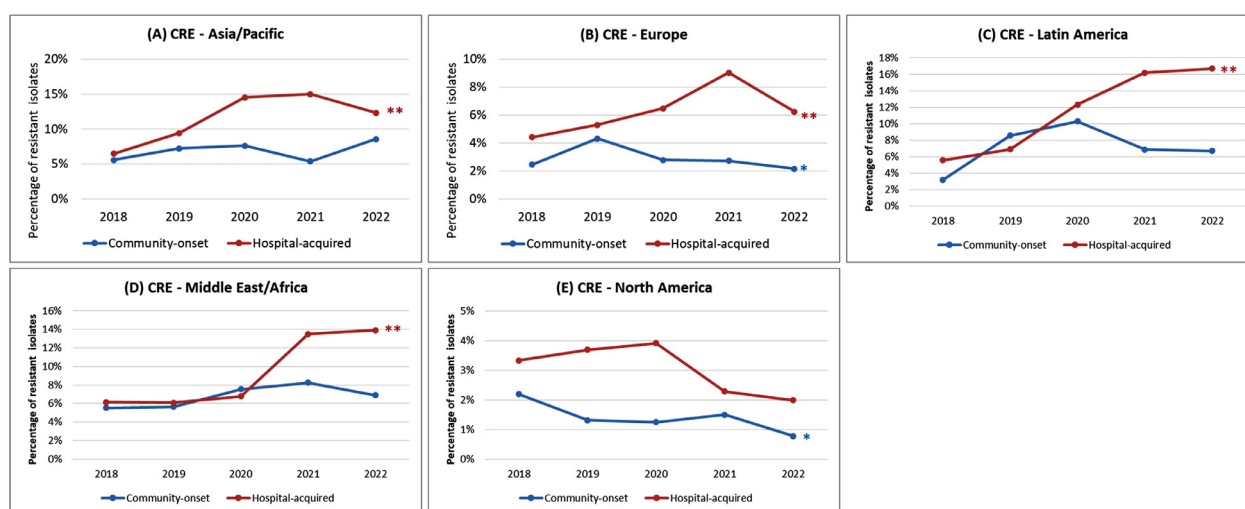


Fig. 3. Longitudinal resistance trends from 2018 to 2022 in the percentage of isolates identified as CRE among the presumed community-onset and hospital-acquired Enterobacterales infections by global region. Significance in trend over time determined by Cochran-Armitage test: ** $P < 0.001$, * $P < 0.05$. Percentages presented in this figure are provided in Supplementary Table S4.

after hospital admission) and hospital-acquired infections (isolated ≥ 48 h after hospital admission) for each region (Figure 3). With the exception of NA, isolates from all regions displayed statistically significant increasing trends of CRE incidence among the hospital-acquired infections. Among isolates from community-onset infections, no statistically significant CRE trends were detected for isolates from the APAC, LATAM, and MEA regions, whereas in EUR and NA, decreasing rates of community-onset CRE identification were observed.

3.4. Regional DTR rates

Enterobacterales with a DTR phenotype increased significantly over the 5-y period in all five regions (Fig. 4). This trend was most pronounced among isolates collected in LATAM where DTR increased from 3.0% in 2018 to 9.9% in 2022 ($P < 0.0001$) and in MEA where DTR increased from 2.7% in 2018 to 8.7% in 2022 ($P < 0.0001$). In the APAC region, as was observed for CRE, the rate

of DTR detection was highest for isolates collected in 2020 (9.8%), and then decreased to 8.8% in 2021 and 9.3% in 2022. Although isolates from both EUR and NA displayed increasing trends over the evaluation period, the rate of increase was less than observed in the other three regions, although in NA a considerable increase from 2021 (2.6%) to 2022 (4.4%) was noted. Among *P. aeruginosa*, no major trends in DTR detection were observed in any region with only isolates from EUR displaying a statistically significant trend (decrease) in DTR detection ($P = 0.0005$). For *A. baumannii-calcoaceticus* complex, the rate of DTR identification was consistently >25 percentage points lower for isolates collected in NA compared to isolates collected in other regions each year. Trends in *A. baumannii-calcoaceticus* complex DTR detection rates were largely absent, although isolates from the MEA region showed a decrease over the evaluation period (from 72.7% in 2018 to 58.8% in 2022; $P = 0.0102$), while isolates from EUR displayed an increase from 45.6% in 2018 to 55.5% in 2021 followed by a decline to 49.4% in 2022 ($P < 0.0001$).

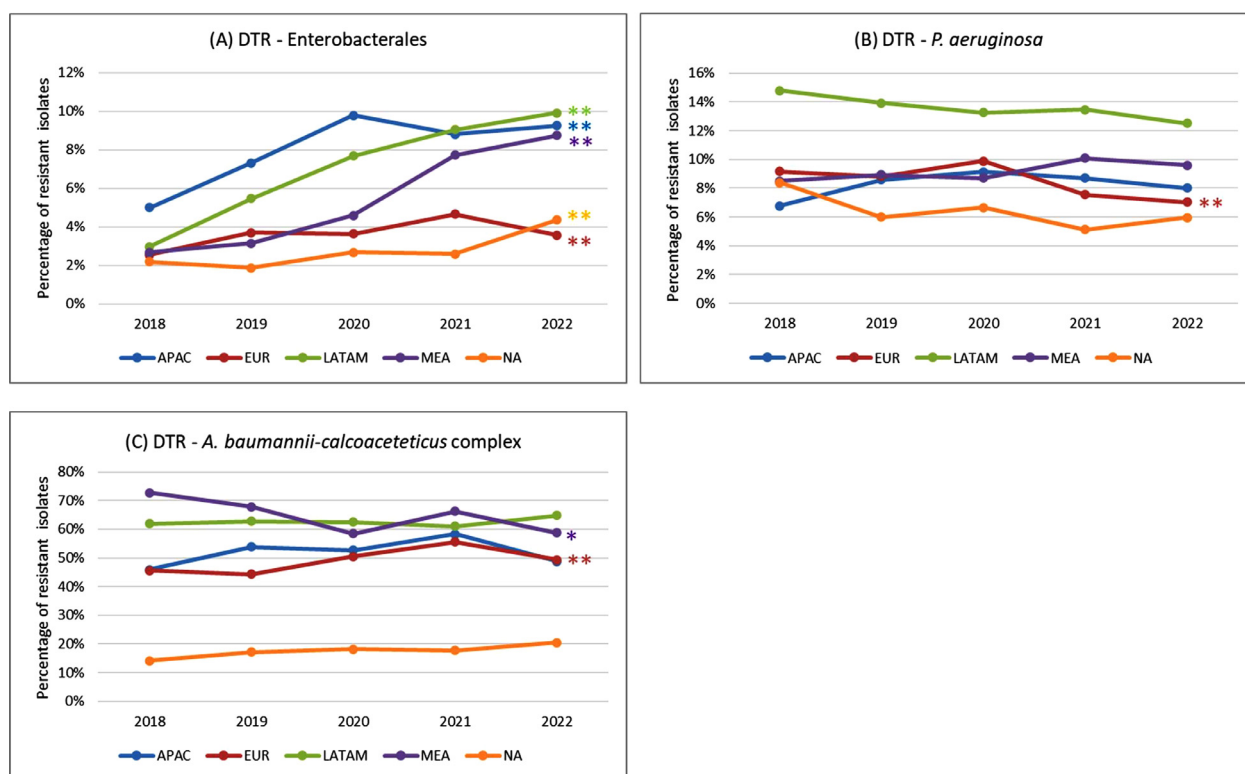


Fig. 4. Longitudinal resistance trends from 2018 to 2022 in the percentage of isolates identified as DTR among all Enterobacterales (Panel A), *P. aeruginosa* (Panel B), and *A. baumannii-calcoaceticus* complex (Panel C) isolates by global region. Significance in trend over time determined by Cochran-Armitage test: ** $P < 0.001$, * $P < 0.05$. APAC, Asia/Pacific; EUR, Europe; LATAM, Latin America; MEA, Middle East/Africa; NA, North America. Percentages presented in this figure are provided in Supplementary Table S5.

3.5. Regional trends in carbapenemase gene carriage

Figure 5 illustrates the proportion of specific carbapenemases detected among meropenem-nonsusceptible Enterobacterales from each region from 2018 to 2022. For the APAC region, the incidence of meropenem nonsusceptibility peaked in 2020, and the three most prevalent carbapenemase genotypes carried by these isolates each year were NDM, OXA-48-like, and organisms co-carrying both NDM and OXA-48-like enzymes. This is in contrast to the isolates collected in EUR, for which KPC was the most frequently identified carbapenemase each year. Among isolates collected in LATAM, where an increasing overall trend of meropenem-nonsusceptibility was clearly observed, isolates carrying KPC and NDM were most numerous each year, with NDM accounting for an increasing percentage annually (with the exception of 2021, in which rates of KPC detection increased and rates of NDM detection declined slightly compared to the previous year). KPC was rare among the isolates from the MEA region, but OXA-48-like enzymes were common and the proportion of meropenem-nonsusceptible Enterobacterales attributable to NDM carriage increased each year. For the isolates collected in NA, meropenem-nonsusceptibility decreased annually. KPC was the carbapenemase most frequently found in meropenem-nonsusceptible Enterobacterales each year, and although still rare, isolates carrying NDM were increasingly observed in 2021 (0.13%) and 2022 (0.28%).

4. Discussion

Data generated by global surveillance programs such as ATLAS can be used to estimate resistance trends over time (generally, year-to-year), however, their results, and comparison to other studies, must be considered in light of: isolate collection protocols; local, national, and regional antimicrobial treatment prac-

tices; as well as antimicrobial stewardship and infection control and prevention policies [12]. The global data generated in the current study was also undoubtedly influenced (starting in early 2020) by the SARS-CoV-2 pandemic. The pandemic has been associated with increased inappropriate antimicrobial use, disrupted routine infection prevention and control practices in healthcare settings, and diverted resources from antimicrobial resistance surveillance and response, all of which may have affected the bacterial infection and antimicrobial resistance rates seen in our study. Considerable heterogeneity exists in antimicrobial resistance trends reported across studies looking at the influence of the SARS-CoV-2 pandemic on antibacterial agent resistance rates [13–16]. Several studies have reported an increase in the rate of CRE infection and colonization, emphasizing the complex and varied effects of the pandemic on antimicrobial resistance patterns [17]. Additionally, the pandemic has been associated with significant increases in healthcare-associated infections, including those caused by CRPA and CRAB [18]. Notably, a retrospective study found that the incidence of CRE colonization in ICU patients increased during the pandemic, underscoring the need for continued surveillance and infection control measures [19]. Taken in total, the observed fluctuations in the rates of carbapenem-resistant pathogens during the pandemic highlight the dynamic nature of antimicrobial resistance and the importance of ongoing surveillance and research to understand and address these changes, mitigate the impact of antimicrobial resistance, and ensure effective treatment and infection control strategies.

In the current study, we only included isolates from medical centres that participated every year from 2018 to 2022 and observed that the annual CRE rate increased significantly in APAC, EUR, LATAM, and MEA regions and decreased in NA from 2018 to 2022 while the DTR rate for Enterobacterales increased significantly in all five global regions (Figures 1 and 4). For all re-

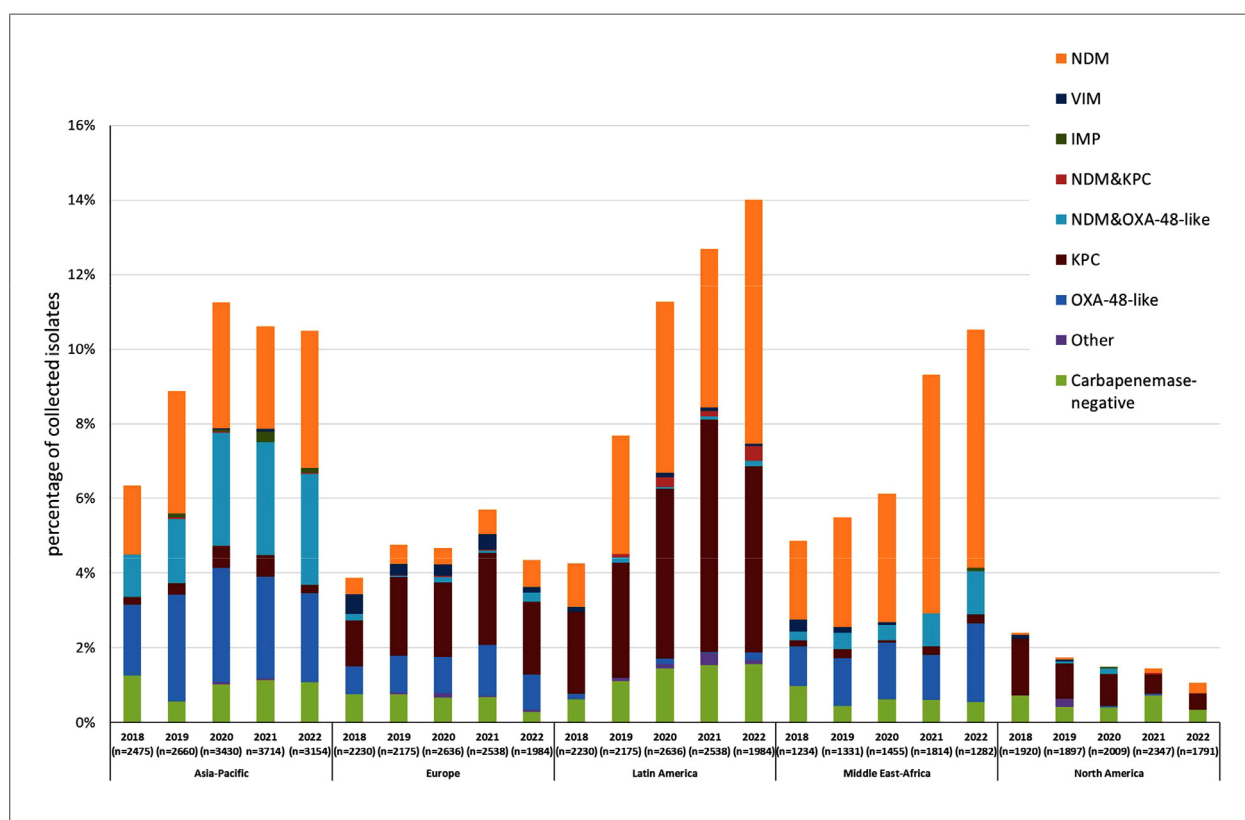


Fig. 5. Proportion of carbapenemase genotypes identified in meropenem-nonsusceptible Enterobacterales collected globally from 2018–2022. 'Other' includes isolates co-carrying NDM&KPC&VIM, NDM&IMP, NDM&VIM, VIM&KPC, KPC&OXA-48-like, VIM&OXA-48-like, or GES variants with reported carbapenemase activity.

gions except NA, the bulk of the change in CRE rates was attributable to increases among pathogens isolated from patients with hospital-acquired infections (Figure 3). Our regional percent CRE data showed that, in 2022, CRE rates exceeded 10% in APAC, LATAM, and MEA regions and agreed approximately with estimated carbapenem-resistant *Klebsiella pneumoniae* rates presented in a recent (2019) comprehensive review of data from 104 countries (APAC, <5–<60%/country; EUR, <5%–<30%/country; LATAM, <5–<50%/country; MEA, <5–<50%/country; and NA, <5%–<10%/country) [1]. Our study also confirmed an earlier report of a trending decrease in CRE detection rates in the United States from 2016 to 2020 [20]. Enterobacterales with a DTR phenotype approached 10% for isolates from APAC, LATAM, and MEA in 2022, rates much higher than reported in earlier publications from those regions [21–23]. Even the percent DTR for NA isolates in 2022 (>4%) is much higher than reported by earlier studies of clinical isolates collected in the United States ($\leq 1\%$) [4,24]. Among meropenem-nonsusceptible Enterobacterales, KPC was the most frequent carbapenemase in EUR and NA each year in agreement with an earlier global study [5], although NDM incidence appears to be increasing (Figure 5) in both regions relative to the earlier study. NDM was the most prevalent carbapenemase detected in 2022 in the other global regions, in contrast to earlier studies that showed KPC to be the most common carbapenemase in LATAM [5]. The emergence of NDM as the predominant carbapenemase in the APAC, LATAM and MEA regions is a grave concern because of the limited number of therapeutic options available to clinicians.

CRPA rates were high in all five global regions. They ranged from 19% (APAC) to 30% (LATAM) in 2022 and remained relatively stable over time (2018–2022) in all five global regions, in agreement with other global surveillance initiatives [25]. In 2022, *P. aeruginosa* DTR rates ranged from 6% (NA) to 12% (LATAM) and remained relatively stable over time in all five global regions, again

in approximate agreement with earlier studies of clinical isolates from APAC, LATAM, MEA, and NA regions [4,21–23,25].

CRAB annual percentage values were very high (50–70% in 2022) in APAC, EUR, LATAM, and MEA regions (Figure 1). The percentage of isolates that were CRAB remained relatively stable over time (2018–2022) in all five global regions. Notably, rates of CRAB and DTR among *A. baumannii-calcoaceticus* complex isolates were consistently >25 percentage points lower in North America than the other regions each year. Our regional percent CRAB values in 2022, agreed approximately with previously published data for all five regions [1] and recently published DTR rates for APAC and NA [4,21].

The IDSA publishes treatment guidance for antimicrobial resistant Gram-negative infections that include CRE, DTR *P. aeruginosa*, and CRAB [26]. Treatment recommendations depend upon the presence or absence of carbapenemases in isolates with CRE, DTR *P. aeruginosa*, and CRAB phenotypes and the identity of the carbapenemase(s), if present. Ceftazidime/avibactam, meropenem/vaborbactam, imipenem/relebactam, sulbactam/durlobactam, aztreonam/avibactam, cefepime/taniborbactam and/or cefiderocol are among the agents with greatest potential utility for the treatment of CRE, DTR *P. aeruginosa* and CRAB infections following pathogen isolation, identification, and determination of in vitro susceptibility.

The relative differences in the percentage of CRE isolates that were community-associated versus hospital-associated varied across different regions. In the APAC, LATAM, MEA, the prevalence of community-onset CRE showed an increasing trend. Conversely, in NA and EUR, the majority of CRE cases were hospital-acquired. The observed variations in the percentage of community-onset versus hospital-acquired CRE isolates underscore the dynamic nature of CRE epidemiology and the need for region-specific infection control strategies. These findings highlight the

importance of tailored interventions to address the evolving landscape of CRE prevalence and transmission in different geographic regions.

Results reported by this study may be subject to certain limitations. In order to not bias our data set, we limited this study to isolates from clinical sites that participated each year of the evaluation period. This reduced the total number of isolates available for analysis by approximately 30%. As an extreme example, 35 hospitals in China contributed isolates to the ATLAS program in at least one study year from 2018 to 2022; however, just one site consistently contributed each year. It is likely that the SARS-CoV-2 pandemic hampered ATLAS participation for some medical centres in 2020 and 2021, as healthcare resources were notably strained. ATLAS is a targeted surveillance program for which each site collects a predefined number of isolates of a given species annually. This is in contrast to other surveillance programs which are more prevalence-based (i.e., collection of consecutive pathogens encountered regardless of species). As such, data from the ATLAS program should not be used for strict epidemiological assessment. Additional limitations of this study include the fact that carbapenemase carriage was only assessed for meropenem-nonsusceptible Enterobacterales isolates (MIC ≥ 2 mg/L) which could have resulted in failure to identify some isolates carrying carbapenemases with low hydrolytic activity (like some OXA-48-type enzymes) that do not consistently confer resistance to meropenem. For this reason, EUCAST guidelines suggest carbapenemase screening for Enterobacterales testing with meropenem MIC values >0.125 mg/L [27].

Antimicrobial resistance among common Gram-negative pathogens is an increasingly important threat to global public health. We conclude that CRE and DTR Enterobacterales are increasing globally and that CRPA, CRAB and their DTR subsets are already at substantial levels and present a barrier to effective treatment in many patients. Current antimicrobial stewardship and infection control and prevention programs, in general, have not reversed increases in Gram-negative antimicrobial resistance globally; hospital-based programs should be strengthened and community-based antimicrobial stewardship and infection control and prevention initiatives should be implemented. Continued development of new and more effective antimicrobial agents, particularly with bactericidal activity against pathogens carrying MBLs, are also a priority. Decisive action is needed to prevent and control the spread of antimicrobial-resistant pathogens. All initiatives to curb antimicrobial resistance must be supported by continued surveillance at local, national, regional, and global levels.

Declarations

Funding: The ATLAS global surveillance program is funded by Pfizer.

Competing interests: MGW and DFS are employees of IHMA, which serves as the central laboratory for the ATLAS global surveillance program. JAK is a consultant to IHMA. MGW, DFS, and JAK do not have a personal financial interest in the sponsor of this manuscript (Pfizer, Inc.). NM, EDH, SK, and AT are employees of Pfizer, Inc. AS has received fees for lectures and serving on advisory boards from Pfizer. LSPM has received fees for lectures, travel support, and participating on advisory boards for Pfizer. DLP has received fees for lectures and travel support from Pfizer. AB has received speaker bureau fees from Pfizer. All authors contributed to the writing, editing, or review of this manuscript.

Ethical approval: Not required.

Acknowledgements

The authors would like to thank all participating investigators and laboratories that contributed to the ATLAS global surveillance program.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2024.03.020](https://doi.org/10.1016/j.jgar.2024.03.020).

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